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Hydrological impacts of rainwater harvesting (RWH) in a case study catchment: The Arvari River, Rajasthan, India
Part 2. Catchment-scale impacts

C.J. Glendenning [a,*], R.W. Vervoort [b]

[a] International Food Policy Research Institute, NASC Complex, CG Block, Pusa, New Delhi 110012, India
[b] Hydrology Research Laboratory, Faculty of Agriculture, Food and Natural Resources, The University of Sydney, Sydney, NSW 2006, Australia

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ABSTRACT

A key question in relation to rainwater harvesting (RWH) is whether the technique increases the sustainability of irrigated agriculture. A conceptual water balance model, based on field data from the Arvari River catchment, was developed to study and understand catchment-scale trade-offs of rainwater harvesting (RWH). The model incorporates an effective representation of RWH function and impact, and works on a daily time step. Catchment spatial variability is captured through sub-basins. Within each sub-basin hydrological response units (HRUs) describe the different land use/soil combinations associated with the case study catchment, including irrigated agriculture. Sustainability indices, based on irrigated agriculture water demand, were used to compare conceptual management scenarios. The results show that as RWH area increases, it reaches a limiting capacity from where additional RWH structures do not increase the benefit to groundwater stores, but reduces stream flow. If the irrigation area is increased at the optimal level of RWH, where the sustainability indices were greatest, the resilience of the system actually decreased. Nevertheless RWH in a system increased the overall sustainability of the water resource for irrigated agriculture, compared to a system without RWH. Also RWH provided a slight buffer in the groundwater store when drought occurred. The conceptual analysis highlights the important link between irrigation area and RWH area, and the impact of RWH on the catchment water balance.
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1. Introduction

Rainwater harvesting (RWH) is a small-scale catchment development tool with a long tradition in many countries, including India (Agarwal and Narain, 1997). In many parts of India RWH captures and stores the intensive monsoonal runoff to recharge groundwater. Where RWH is present, it increases the capacity of small-holder farmers to irrigate with groundwater, so there is a greater need to understand how this catchment development tool impacts larger scale catchment water balances. The resulting shifts in the water balance could impact the resilience of agricultural livelihoods, which means communities could be more vulnerable to shocks such as drought (Mudrakartha, 2007). An individual RWH structure may not have a large effect on stream flow or aquifer storage, but the cumulative effect of many RWH structures within a large catchment would be more significant. For example the construction of farm dams in some catchments of the Murray Darling Basin in Australia caused statistically significant reductions in the quantity of potential stream flow response (Schreider et al., 2002).

Extensive detailed field studies could provide some answers, but local-scale measurement of recharge is difficult as most field methods are expensive, time consuming, or do not deliver the desired accuracy (Healy and Cook, 2002). In addition high temporal and spatial variability of rainfall, soil and aquifer hydraulic properties means that long time series of data are needed (Silberstein, 2006). However, data is very limited in semi-arid/arid regions where populations are sparse and economic resources are limited (particularly in developing countries) (Sen, 2008). Given the general lack of hydrological field data in semi-arid catchments, a model is a useful and inexpensive way of extrapolating limited field data and investigating management scenarios in a larger scale catchment. Simulation modelling of catchment hydrology might offer a way of assessing the impact of RWH at a larger catchment scale, but model input and accuracy will determine the usefulness of the final results (Silberstein, 2006). Specifically, this study uses a conceptual model to better understand the processes and dynamics of RWH to guide further detailed investigations.

The semi-arid areas where RWH is found means that the large data requirements of existing models, for example the Soil Water Assessment Tool (SWAT) (Arnold and Fohrer, 2005), cannot be met. A key limitation is the lack of stream flow gauging data, which means calibration and validation of a hydrological model

* Corresponding author. Tel.: +91 011 2584 6565; fax: +91 011 2584 8008.
E-mail address: c.glendenning@cgiar.org (C.J. Glendenning).

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is almost impossible. Predictions in semi-arid areas are also difficult because climatic conditions tend to be temporally and spatially variable (Wheater and Al-Weshah, 2002). Additionally, the surface water–groundwater interactions, which are necessary to represent RWH hydrological processes, do not exist in many models. For example the Tool for Estimating Dam Impact (TEDI) model, used to estimate small farm dam impacts in Australia, has no groundwater component.

The aim of this paper is to describe the development and application of a conceptual model to study trade-offs related to RWH impacts in a catchment from a sustainability perspective. The model will be a generalised conceptual system based on the field scale impacts of RWH in the semi-arid Arvari River catchment, described in the companion paper (Glendenning and Vervoort, 2010). The model is applied to study the sustainability of irrigated agriculture with and without RWH in a typical catchment.

1.1. Water resources sustainability indices

Due to collection and storage of runoff, RWH may change the catchment water balance, but also may increase the sustainability of irrigated agriculture as a result of more reliable groundwater stores. To quantify that impact, sustainability indices are a useful way to measure changes in a water resource system. Quantification of the sustainability of water resource systems can be achieved by using measures of reliability, resilience and vulnerability. The indices are based on whether a specified demand threshold is met by a defined water resource system (Ajami et al., 2008; Hashimoto et al., 1982; Loucks, 1997; McMahon et al., 2006; Zongxue et al., 1998).

Hashimoto et al. (1982) describe three criteria for evaluating the performance of water resource systems:

1. Reliability (*RE*) is the frequency or probability that a system is in a satisfactory state (Fowler et al., 2003):

$$RE = prob[X_t \in S] \quad \text{where } RE = \sum_{t=1}^{T} Z_t \tag{1}$$

X_t is the output state or status of the system at time t, Z_t is a binary measure where X_t is either an element of S values (output or performance of the water resource system in a satisfactory state) or F values (output or performance of the water resource system in a failure state). The *RE* index ranges from 0 to 1, where 1 reflects 100% reliability.

2. Resiliency (*RS*) is the probability that when a system is in a failure state, the next time step is a satisfactory state (Ajami et al., 2008; Fowler et al., 2003):

$$RS = prob\{X_{t+1} \in F\} \quad \text{where } RS = \sum_{t=1}^{T} \frac{W_t}{T - \sum_{t=1}^{T} Z_t} \tag{2}$$

$W_t = 1$ if X_t is element of F and then X_{t+1} is an element of S otherwise $W_t = 0$. The resilience index ranges from 0 to 1, where 1 is a system with 100% resilience.

3. Vulnerability (*V*) is the maximum of the sum of the difference between the threshold (criteria C) and the actual level (X_t) for any failure periods (J_i) (Ajami et al., 2008; Fowler et al., 2003).

$$V = \max\left\{\sum t \in Ji\, C - X_t, \quad i = 1, \ldots, N\right\} \tag{3}$$

It thus reflects the severity of the failure. The vulnerability index can have a wide range depending on the difference between X_t and C, but higher values indicate higher vulnerability, lower values represent a less vulnerable system.

In order to quantify these indices, the operational status of the water resource system needs to be described as either satisfactory (S) or unsatisfactory (F), using defined thresholds. The thresholds for the calculation of the sustainability indices in this study relate to water demand for irrigated agriculture. There is sufficient evidence to show that there is a direct link between irrigation and poverty alleviation. Groundwater development increases irrigation area, which leads to agricultural growth (Mukherji and Shah, 2005) and is seen as a more effective way to target poverty than surface water, due to its reliability and spatial availability (Wegerich, 2006). Additionally, as RWH increases groundwater recharge, where RWH is introduced, irrigated agriculture increases (Srivastava et al., 2008). Consequently, the sustainability indices provide a useful method to compare differences in catchment water balances, specifically the groundwater resource, under different land use scenarios, including the area of RWH.

2. Study area description

The catchment chosen to base the modelling work is the semi-arid ephemeral Arvari River (476 km^2), located in the eastern part of the Indian state of Rajasthan. Since 1985, over 366 rainwater harvesting structures have been built throughout the catchment by the community and a local non-government organisation (NGO), Tarun Bharat Sangh. The different RWH structure types include:

- *ANICUT*: small concrete dam or weir that dams the main reach of the river.
- *BANDH*: small concrete dam that dams the tributary to the main river reach.
- *JOHAD*: small earthen crescent moon shaped dam found at the foothills of small hilly catchments.
- *TALAB*: village ponds for bathing and livestock drinking, similar to *Johads*.

Data is sparse; the catchment is ungauged with no climate station, and existing information on aquifer properties, geology and soils is at a very large scale, which does not capture the spatial and temporal variability at finer scales within the catchment. Eighty percent of the rain falls during the south-west monsoon (June-September). The closest rainfall station to the Arvari River catchment is in Thanagazi (15 km to the North of the catchment), which has an average yearly rainfall of 705.8 mm and 8 rainfall days greater than 35 mm with an average of 31 rainy days a year (1901–2002). There is limited small-scale information about geological and soil characteristics in this catchment, though field soil samples revealed land use reflects soil type. Field-scale impact studies of RWH were carried out in 2007–2008 with estimates of RWH recharge and rainfall characteristics (Glendenning and Vervoort, 2010).

3. Methods

3.1. The model

An existing highly parameterised model would not have been suitable to model the hydrological processes in the Arvari River catchment because of the large gaps of knowledge and data, including lack of data on aquifer properties and basic hydrological data, including long term weather observations in the catchment and a gauging station on the river. In contrast, simpler existing models do not incorporate the hydrological processes related to RWH. As a result a conceptual water balance model was purpose built to capture the relevant hydrological processes in the catchment that are influenced by RWH. This includes surface water–groundwater interactions and recharge volumes from RWH.

The model developed in this study is a balance between a relatively realistic representation of a semi-arid catchment with RWH and the limited data available for such studies. As a result the final model remains conceptual but has realistic field values as input. The daily time step model was developed in R (R Development Core Team, 2007) based on field information from the study of the Arvari River catchment (Glendenning and Vervoort, 2010).

3.1.1. Sub-basins

The Arvari River catchment displays great variability in climatic and landscape conditions (Glendenning and Vervoort, 2010). To capture some of the variability, the modelled catchment was divided into three smaller units or sub-basins. One main river reach is divided into sub-basin 1 and sub-basin 2, while another river reach is sub-basin 3. Groundwater and surface water catchments are assumed to align, which oversimplifies the physical scenario, but there is insufficient data to determine accurately the physical situation.

Based on the local hydrogeological information an upper shallow alluvial aquifer, which is hydraulically connected to a deeper aquifer (Gale et al., 2002), was included in each sub-basin. In the model, the water that enters the deeper aquifer is considered lost from the system. The key determinant for defining the shallow aquifer storage limits was to capture the fluctuations in the observations in the catchment wells (Glendenning and Vervoort, 2010). As no baseflow was observed in the field over 2 years, the maximum storage in the aquifers was set sufficiently high to prevent baseflow in the model. The initial water level in the shallow aquifer was based on the average groundwater depth of the monitored dug wells on 1/1/2008, which was approximately half of the maximum storage capacity.

The groundwater balance for each shallow aquifer in each sub-basin is calculated using recharge from different land uses (R), extraction for irrigation when present (Irr), lateral flow between sub-basins (L) and loss to the deeper aquifer (Pc) (Eq. (4)).

$$\Delta GW = R - Irr - L - Pc \quad (4)$$

ΔGW = Change in shallow aquifer storage

Water lost to the deeper aquifer was calculated using a percolation coefficient, which determines the fraction of water from the shallow aquifer lost to the deeper aquifer (Arnold et al., 1993). There are no field estimates and given the conceptual nature of the model, the percolation coefficient was assumed to be quite small and set to an arbitrary value of 0.001 mm/day. As a result of this, the filling of the shallow aquifer could be over-estimated. However, as the connection between the upper and lower aquifer is unknown the size of the shallow aquifer was used as the control variable. Greater values of the percolation coefficient would result in the model estimating smaller sizes of the shallow aquifer.

Water in the shallow aquifer moves laterally between sub-basins (Q_L) based on Darcy's Law and is described as (Anderies, 2005):

$$O_L = K \cdot A \frac{(E1 + D2 - D1)}{D3} \quad (5)$$

K is hydraulic conductivity (m/day), A cross-sectional area (m^2), $D3$ length of flow path connecting the aquifer in the upper sub-basin with the aquifer in the lower sub-basin (m), $E1$ elevation of upper sub-basin 1 relative to the lower sub-basin 2 (m), i.e. the difference in elevations, $D1$ depth to the water table relative to the elevation in the upper sub-basin 1 (m), which fluctuates daily, $D2$ depth to water table relative to the elevation in the lower sub-basin 2 (m), which also fluctuates daily.

Due to limited information, K, A and $D3$ could not be defined, so Q_L was estimated to be 0.0001 mm/day. Flow from the shallow aquifer of sub-basin 1 moves into the shallow aquifer of sub-basin 2. However, lateral flow from the shallow aquifer in sub-basin 2 and sub-basin 3 occurs below the modelled catchment. The percolation coefficient in sub-basin 2 and sub-basin 3 was therefore increased to include both movement of water into a deeper aquifer and water moving laterally downstream.

Recharge from the surface enters the shallow aquifer one day after the calculated actual recharge, based on the cross correlation analysis between potential recharge from RWH and groundwater levels from in the field (Glendenning and Vervoort, 2010).

Table 1
Statistical summary of Thanagazi rainfall (1980–2006) compared with the measured rainfall (P) from the Arvari River catchment at Bhaonta and Hamirpur villages in 2007 and 2008. Wet P represents the rainfall during the monsoon months (July–October), while Dry P represents the rainfall during the remaining dry months.

Station	Annual mean			Annual standard deviation			Annual log based mean		
	P	Wet P	Dry P	P	Wet P	Dry P	P	Wet P	Dry P
Thanagazi	695	599	96	271	259	73	636	535	0
Bhaonta	598	554	54	335	259	76	549	512	0
Hamirpur	696	662	35	351	301	50	651	626	0

3.1.2. Rainfall

Rainfall is both spatially and temporally variable. Given the conceptual nature of the modelling, it was decided not to use the short time series of observed daily data collected in the catchment over 2 years, or the long term annual data from the Alwar station (43 km from the catchment). Instead a stochastic model to simulate the rainfall was used to create a 12 year time series of daily realisations of rainfall for each sub-basin. In this model, rainfall is defined as a Poisson process in time with rate λ (day^{-1}) defining the exponential distribution of times between rainfall events (Fernandez-Illescas and Rodriguez-Iturbe, 2004; Laio et al., 2001a). The mean depth of rainfall events is subsequently defined by the parameter α (mm), also based on an exponential distribution of rainfall depths.

The parameters α and λ were calculated from the daily rainfall data at the Thanagazi station (1980–2006) (15 km from the catchment) and the annual rainfall at the Alwar station (1901–2002). The rainfall at Thanagazi indicates similar properties to that of the field results, but covers a longer time period (Table 1). As the monsoon season is the most significant period of rainfall, the parameters were calculated separately for the monsoon months (June–September), and the drier months (October–May) (Table 1).

Rainfall is also variable between years which is influenced by the variation in monsoon seasons. To predict this inter-annual variation, the α value for the monsoon seasons was varied between years. As a first step, a smooth function (based on a Fourier series approach) was fitted to the annual rainfall series at the Alwar station. However predictions based on this function underestimated the observed distribution of annual rainfall (Fig. 1a). A random component with the standard deviation of the Alwar annual rainfall distribution was therefore added to match the observed variability (Fig. 1a). The subsequently modelled and observed annual rainfall are similar (Fig. 1b), with an observed mean of 618 mm, and predicted mean of 609 mm, and observed variance of 29,983 mm^2 and predicted variance of 37,315 mm^2. Fig. 1c shows that the observed annual rainfall from the field data falls within the modelled distribution. While the generated rainfall describes the average and follows similar distribution to the field results (Glendenning and Vervoort, 2010), the intra-sub-basin variability is not captured in this model, as the field data demonstrated high spatial variability. Therefore in the conceptual model, for each sub-basin, rainfall was subsequently generated independently using the same model parameters. This resulted in different rainfall events in each of the

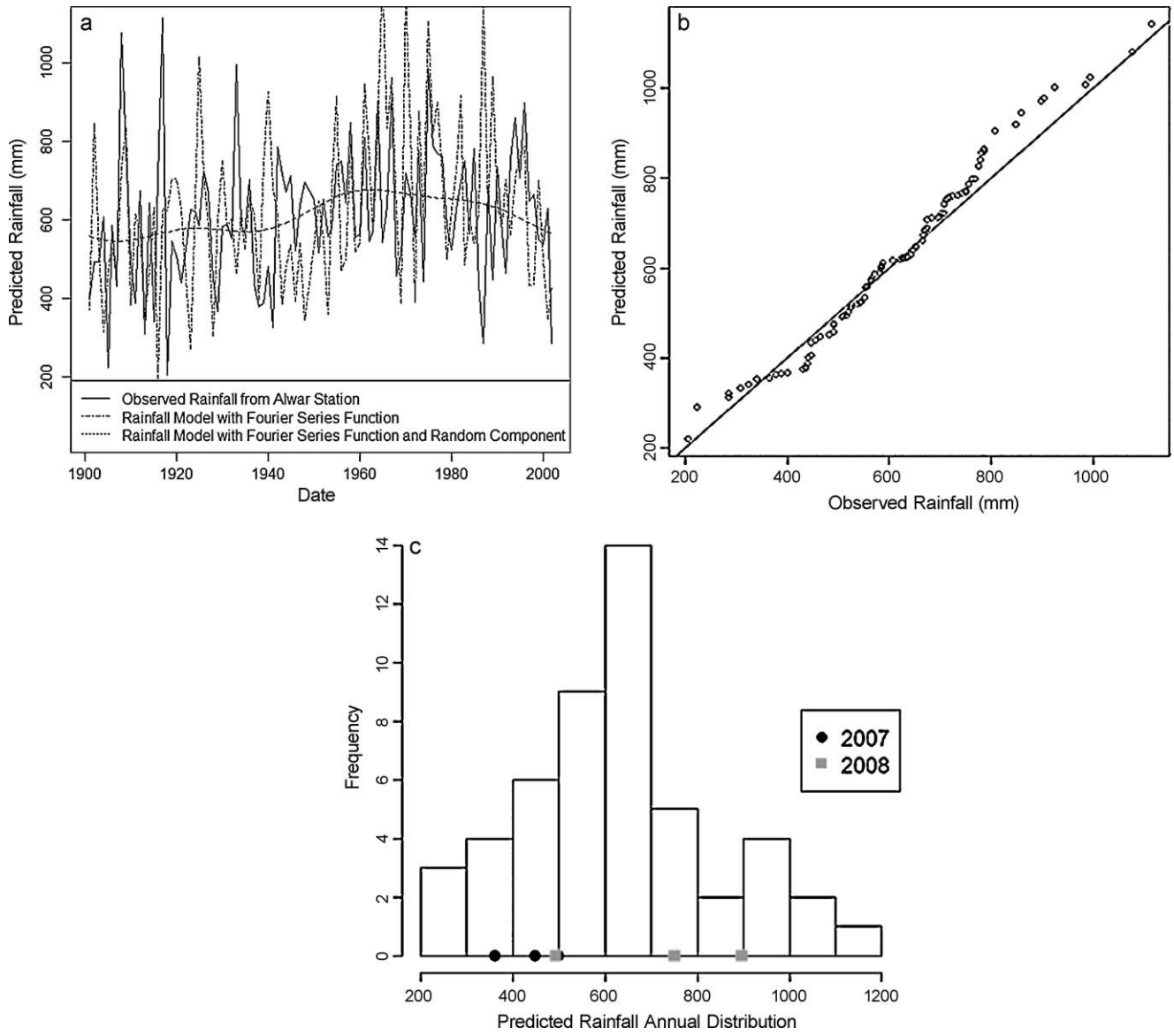


Fig. 1. (a) Modelled annual rainfall using the Fourier Series function and with a random component compared with Alwar annual rainfall. (b) Quantile–quantile plot of modelled annual rainfall against observed annual rainfall (mm) (with line $x = y$). (c) Example of one modelled rainfall distribution for 10 years. The points represent collected rainfall during the field study from the three gauges in the catchment in 2007 and 2008.

sub-basins, similar to the field data (Glendenning and Vervoort, 2010). More complex rainfall generating models could be used but would detract from the main model processes related to RWH.

3.1.3. Evapotranspiration

Evaporation for RWH structures is set to the potential evaporation (*PotET*). Potential evaporation was predicted from the daily *PotET* from Alwar district data (India Water Portal, 2009), using a similar Fourier series approach that was used to predict annual rainfall. Actual evapotranspiration (*ET*) was based on a piece wise linear function, which has been widely used in ecohydrological water balance models (Fernandez-Illescas and Rodriguez-Iturbe, 2004; Laio et al., 2001b; Teuling and Troch, 2005). In this calculation of actual *ET*, water loss is mainly governed by the soil moisture status. From maximum *ET*, actual *ET* losses decrease linearly from AW_{cr} (critical available water, where plants begin to close stomata and soil water availability becomes important) to *WP* (wilting point) (Laio et al., 2001b):

$$ET = fr \cdot \beta(1 - \exp(c \cdot LAI)) \cdot PotEt \tag{6}$$

The parameter c is the light use efficiency parameter, which was set to 0.4 (Slavich et al., 1999), β is the soil moisture stress function, based on available water content (*AW* in mm) given by:

$$\beta = \begin{cases} 0 & \text{for } AW \le WP \\ \dfrac{AW - WP}{AW_{cr} - WP} & \text{for } WP < AW \le AW_{cr} \\ 1 & \text{for } AW_{cr} < AW \le FC \end{cases} \tag{7}$$

where *FC* is the field capacity. The transition between unstressed and stressed transpiration is defined by critical available water (AW_{cr}), and the wilting point (*WP*) is when plants can no longer extract water from the soil matrix (Teuling and Troch, 2005). The parameter f_r represents the root fraction in the soil (Teuling and Troch, 2005), which was assumed to be homogeneous and set to 1. In a later section the derivation of *WP* and AW_{cr} is explained. Leaf Area Index (*LAI*) is defined for each land use. For the agricultural land use, depending on the number of cropping seasons within the year, *LAI* increases to a maximum of 5 and declines to 0.1 (Table 2). This reflects literature estimates where in China, Li et al. (2003) found wheat had a maximum *LAI* of 5. Rastogi et al.

Table 2
Description and Leaf Area Index (*LAI*) for each land use in the conceptual model.

Land use	Description	Month	*LAI*	Landscape position
Agriculture	Ag1 Rain-fed *Kharif*	July–September October–June	4 0.4	2–3 km from river. Relies on rainfall to increase soil moisture content.
	Ag2 Irrigated *Kharif* and *Rabi*	July–September, November October, December–January February–June	4 5 0.1	Along river. Return flow from groundwater irrigation.
	Ag3 Irrigated *Kharif*, *Rabi*, *Zaid*	July–September, November, March–June October, December–February	4 5	
Commons	Sparse forest/grazing/wasteland	July–January February–June	4 0.5	Steep slopes. High runoff, exposed, 1–3 m trees in leaf during monsoon.
RWH	*Bandhs* (includes *Anicuts*)			Upper catchment, near/in river
	Johads (includes *Talabs*)			Lower hills

(2000) also observed a maximum wheat *LAI* of 5 in India, and Tyagi et al. (2003) found the peak *LAI* for maize in a semi-arid area of India to be between 5 and 5.5.

3.1.4. Hydrological response units

Within each sub-basin hydrological response units (HRU) are defined, which represent a land use with unique management factors and soil type and have no spatial interpretation (Arnold and Fohrer, 2005). The water balance is calculated for each HRU in each sub-basin. The partitioning of rainfall into runoff uses the USDA-SCS Curve Number method (USDA-SCS, 1985). All runoff is first routed into the RWH reservoirs, where a maximum depth for RWH is defined. Any water that overflows from the RWH reservoir is moved into the stream, which is then routed into the next sub-basin, subject to transmission losses.

The dominant land uses in the Arvari River catchment were incorporated into the conceptual model. Current land use in the Arvari River catchment closely reflects elevation and soil distribution. In the hills, with higher elevations and shallow rocky soils, land is usually used for grazing with thinly covered forest areas or open land known as commons. In the plains along the river, where the soils are deeper and richer in clay content, land use is predominantly agriculture, including irrigated agriculture. The main cropping seasons are the *Kharif*, the monsoon crop, and *Rabi*, the winter crop. *Zaid* is the summer crop season between April-June. Further from the river, agriculture still persists, but there is less irrigation and the area for *Rabi* and particularly *Zaid* crops is reduced. The main land uses therefore used in the conceptual model are agriculture, commons and RWH (Table 2). The agriculture land use is divided into three management practices: Ag1 represents rainfed *Kharif* crop area only, while Ag2 represents the rainfed *Kharif*, which supplements with groundwater irrigation, and irrigated *Rabi*. Ag3 extends the Ag2 practice to also include irrigated *Zaid*. Domestic and livestock water use was not considered to be significant. The HRU for RWH was divided into two to capture the differences in recharge between the larger and smaller structures, where the *Bandhs* HRU represent *Bandhs* and *Anicuts* and the *Johads* HRU represent *Johads* and *Talabs*.

The fractional area of the total catchment for each HRU area is defined, where the approximate area of the commons HRU is 50%, Ag1 HRU is 35%, Ag2 HRU is 15%, and Ag3 HRU is 1.5%. The RWH fractional area is 1.44% based on local information from the NGO, TBS. In the scenarios, when irrigated agriculture or RWH area is increased, the area is taken from the commons HRU.

Rainfall, irrigation and *ET* influence the soil layer, which represents the layer where root activity takes place. Input into the root layer is Rainfall (P) – Runoff (RO). Recharge of groundwater occurs when the soil layer reaches field capacity (FC), and all excess input water becomes recharge. Irrigation is only included for the Ag2 and Ag3 land uses. Irrigation water is taken from the shallow aquifer in each sub-basin and takes place when the soil profile is at AW_{cr} (Allen et al., 1998), to a maximum daily depth of 150 mm and minimum irrigation depth of 100 mm. These depths were based on field observations of irrigation practices in the study area.

The general simulation approach for the commons land use is similar to the agricultural land use, but no irrigation is included, and a different curve number, *LAI* distribution (Table 2) and soil type is used.

Runoff enters the RWH storage and is multiplied by the fractional area of each contributing land use and then divided by the fractional area of RWH structures in the sub-basin to represent the appropriate depth of water entering the storage areas. The total volume of RWH is considered as one large reservoir at the end of each sub-basin (Loukas et al., 2007). There are different parameter values for the two RWH HRUs, *Bandhs* and *Johads*, which was based on field observations for each structure type (Glendenning and Vervoort, 2010). A soil layer is not simulated in the RWH structures. Instead recharge depth (R in mm) is calculated using Darcy's Law, using change in groundwater height Δh, saturated hydraulic conductivity (K_{sat}) and depth from the surface to the aquifer (l) (Eq. (8)):

$$R = -K_{sat}\left(\frac{\Delta h}{l}\right) \tag{8}$$

A pedotransfer function developed for India was used to calculate K_{sat} (Adhikary et al., 2008) where:

$$K_{sat} = 173.4(silt\% + clay\%)^{-1.48} \tag{9}$$

Using sampled soil textures from RWH storages in the field, the resulting K_{sat} from this pedotransfer function (1.53 m/day) was used for both *Bandhs* and *Johads*. However, *Johads* had a deeper depositional silt layer, which reduced potential recharge in the field (Glendenning and Vervoort, 2010). This was simulated in the model by changing the parameter describing the depth to the shallow aquifer.

Depth to the shallow aquifer was assumed to be 18 m for *Johads*, and 4 m for *Bandhs*, based on the average depth of water levels from the surface in wells in the field close to monitored structures (Glendenning and Vervoort, 2010). *Bandhs*, which also incorporate the *Anicuts* on the river, are therefore much closer to the groundwater table than *Johads*. The maximum depth of each structure is also defined differently; storage in *Bandhs* is 2 m deep, while *Johads* are 1 m. This is approximately half the actual average maximum depth in the field but because the actual storage shapes are not square tubs as represented in the model, this is still representative. Any water above the maximum depth becomes overflow and is moved into the river. The maximum depth also influences the non-linearity of potential recharge where, as rainfall increases, the structure reaches a limiting capacity to induce more potential recharge. In order to interpret this effect in the model, an additional parameter was introduced. Beyond a defined threshold of recharge the following empirical relationship was used to calculate recharge:

$$R = T - T^a + K_{sat}\left(\frac{RW\ \text{storage} + \text{depth to aquifer}}{\text{depth to aquifer}}\right)^a \quad (10)$$

where a is a parameter set to 0.3, while the threshold (T) was fixed at 100 mm/day. All potential recharge from RWH goes directly into the shallow aquifer in each sub-basin.

3.1.5. Soils

To represent realistic soil values in the conceptual model, the two main soils used in the model were based on soils described by the National Bureau of Soil Survey and Land Use Planning in the Arvari River catchment area (Shyampura and Sehgal, 1995). All horizons of the described soils were incorporated into one layer in the model. The Bansur series soil was used for the agricultural land uses, while the Kanwat series is used for the commons land use. Using these soil descriptions, the commons land use soil is much shallower and has a smaller water holding capacity (100 mm) than the agricultural soil, which has a field capacity of 261.2 mm.

Based on the data, *WP* was defined as a fraction of *FC*, to reduce the number of parameters. As a result *WP* was defined as 0.454 *FC* for the agricultural land uses and 0.54 *FC* for the commons land use. The value of AW_{cr}, representing the point at which stomata close, was more difficult to define and is essentially a free parameter. In the literature this parameter varies between 44 and 94% of *FC* (Laio et al., 2001a; Teuling and Troch, 2005). In the model AW_{cr} was defined as 0.7 *FC* for the agricultural land use and 0.78 *FC* for the commons land use. Overall this means that the agricultural soil has a wider range of water availability and plants take longer to reach stress levels than in the commons soil.

3.1.6. Interception and runoff

The USDA-SCS Curve Number (*CN*) was used to estimate runoff from each land use. The *CN* method predicts runoff from rainfall, using a curve number, which represents potential maximum soil retention and reflects land use, antecedent moisture condition and soils (Ponce and Hawkins, 1996). On larger temporal scales, initial losses increase due to transmission and evaporation losses (Simanton et al., 1996). However on a daily time step initial abstraction (I_a) is small. Therefore I_a was changed from 0.3 to 0.05 (Woodward et al., 2003). Antecedent moisture conditions 2 and hydrologic soil group B (medium textured soils) were used to find the *CN* for each land use. The *CN* for each agriculture HRU and commons HRU was defined from the SCS tables (USDA-SCS, 1985), which are similar to a study for south-eastern Rajasthan, where weighted *CN* values ranged from 80 to 83 (Gupta et al., 1997). However the *CN* is a calibration parameter in the model and might increase or decrease for real catchments. The agriculture land uses have two curve numbers, a *CN* of 77 during the time when land is fallow and a lower *CN* of 72 when cropped. The commons land use *CN* was set to 79.

3.1.7. Routing and transmission losses

There are many modelling methods to route stream flow, but with little information on channel characteristics and flow, linear routing was chosen:

$$S = K_Q \quad (11)$$

K is an empirical parameter, describing the storage ($S\,m^3$) in the river reach related to discharge ($Q\,m^3/s$).

In the model the routing coefficient, k, was set to 5. This value indicates little spreading of the flood hydrograph, which would be typical for a semi-arid ephemeral river, such as the Arvari River. In Australia, Costelloe et al. (2003) set k to 4.5 in semi-arid channels in South-western Queensland.

Transmission losses are an important hydrological component of a semi-arid river system (Pilgrim et al., 1988). However due to the limited data on stream flow, the model incorporates regressions developed by Sharma and Murthy (1994), which relate transmission loss to channel characteristics and flow volume in an arid region of north-west India. The formula for calculating transmission losses for the first km is:

$$V_1 = 1.983\,V_{up}^{0.73} \quad (12)$$

V_1 is the transmission loss for first km (m^3), V_{up} inflow volume at the upstream site (m^3).

In the model, this equation was applied to every km of the stream reach consecutively and the resulting transmission losses were summed over the total reach length.

Transmission losses are added to the shallow groundwater. Because the model works in depth only, the transmission loss in each sub-basin calculated using Eq. (12) was divided by the river length and catchment area from a GIS analysis of the Arvari River catchment.

3.1.8. Limited sensitivity analysis

The model has 23 input parameters (Table 3). To better understand the response of the model to the input parameters, a limited sensitivity analysis was performed with the parameters considered to have the highest impact on the output of the model (Lenhart et al., 2002). The parameters *CN*, I_a, *LAI*, *FC*, percolation coefficient, irrigation trigger, AW_{cr}, and *WP*, were considered to have the highest impact. Additionally, to understand how the spatial and temporal variability of rainfall might affect the output, rainfall was included in the sensitivity analysis. The sums of *Johad* recharge and *Bandh* recharge, shallow aquifer water levels, runoff, *ET* and stream flow were used as output variables. For each output a Relative Sensitivity Index was calculated (Lenhart et al., 2002; Ouessar et al., 2008).

3.1.9. Adjustment to realistic field values

A qualitative adjustment of the model to realistic values was based on the 2 years of field data and the resulting observations and conclusions (Glendenning and Vervoort, 2010). This adjustment was aimed to improve the representation of the main processes related to RWH in a semi-arid catchment, in this case the Arvari River catchment. As such, the adjustment brings the model in the realm of possible real catchments rather than purely theoretical.

It was observed that local farmers increase the area of irrigated agriculture based on the groundwater availability, which they judge from the water levels in their dug wells. The actual groundwater use is a complex function, which relates to how the farmers adjust the irrigation area to groundwater availability. In the model, a simplified step-wise function was introduced to mimic this behaviour. If the shallow aquifer level is higher than 125 mm,

Table 3
Model parameters and values.

Component of the model	Parameter name	Value
Groundwater	Percolation coefficient	0.001
	Initial value	100 mm
	Maximum value	200 mm
	Elevation of sub-basin 1 (*E1*)	
	*D*2	Groundwater level and depth to surface in sub-basin 2
	*D*1	Groundwater level and depth to surface in sub-basin 1
Rainfall (*P*)	Rainfall depth α	Varies with each year
	Rainfall length λ	0.5 in monsoon season, 0.06 in dry months
Evapotranspiration (*ET*)	Critical available water AW_{cr}	0.7 *FC* Agriculture, 0.78 *FC* for Commons
	Wilting point (*WP*)	0.454 *FC* for Agriculture, 0.54 *FC* Commons
	Leaf Area Index (*LAI*)	Table 2
	Light use efficiency parameter (c)	0.4
	Soil moisture stress function (β)	(Eq. (6))
	Root fraction (f_r)	1
RWH HRU	Saturated hydraulic conductivity K_{sat}	1534.28 mm/day
	Depth to groundwater table (L)	18000 mm *Johads*, 4000 mm *Bandhs*
	Maximum storage depth	1000 mm *Johads*, 2000 mm *Bandhs*
	Threshold where recharge volume changes	100
	Shape parameter (a)	0.3
HRU's (Ag1, Ag2, Ag3, Commons)	Curve Number (*CN*)	Table 5
	Field capacity (*FC*)	Agriculture 261 mm, Commons 100 mm,
	Irrigation trigger (for Ag2, Ag3)	When soil profile is AW_{cr}

then the defined fractional area of Ag2 and Ag3 increases 1.2 times from the original assigned values.

During the calibration runs, the same rainfall distribution input was applied to all sub-basins and also the fractions of land use were kept constant (which includes the adjustment of the irrigated fraction as described above). RWH area was taken to be 1.44% of the catchment area, which was the approximate level of RWH in the Arvari River catchment in 2005. For RWH recharge, the parameters that were calibrated were the distance to the groundwater (Eq. (8)) and the depth of the structures at which overflow takes place. From the field analysis, *Bandhs* had higher recharge than *Johads*. Transmission losses on the river above *Anicuts* are taken into consideration in the *Bandh* HRU, so *Bandhs* were calibrated to have greater recharge.

Background natural recharge estimates for the commons HRU were based on existing literature. A regression equation based on tracer studies using tritium for alluvium provinces in India, where:

$$R = 0.147P - 6 \quad (13)$$

in which R and P are in mm/year, was used to estimate natural recharge (Rangarajan and Athavale, 2000). Using the rainfall at Hamirpur (Glendenning and Vervoort, 2010), this gives an approximate natural recharge value of 60 mm in 2007, and 126 mm in 2008. The soil water holding capacity of the commons HRU was doubled by increasing the *FC* to match these recharge estimates. The Ag2 and Ag3 HRUs are irrigated so recharge from these units would be higher than the commons HRU as there would be return flow from irrigation events.

The modelled response of groundwater levels in the shallow aquifer was adjusted to reflect the observed water levels (Glendenning and Vervoort, 2010) by changing the size of the shallow aquifer. The groundwater fluctuation observed in that study in the dug wells was multiplied by the estimated specific yield of 0.1, which gave a rise of approximately between 60 and 300 mm/year depending on rainfall and the distance from RWH structures.

3.2. Simulation analysis

To meet crop water demands in the Arvari River, the farmers rely on a combination of groundwater and rainfall in *Kharif* and groundwater in *Rabi*. *Kharif*, the summer or monsoon crop, which is mostly maize, is planted after the first rains in July and harvested in October. *Rabi*, the winter crop, mainly wheat, is planted in November and harvested around March. The water resource system was therefore based on the availability of water in the sowing months of the *Kharif* (July) and *Rabi* (November) crops. The daily available water in the system is defined as the groundwater storage combined with any rainfall. Because the initial available water at the start of each season strongly determines water availability for the rest of the season and the amount of irrigated area planted, any impact of RWH is best based on the first month of each season (July and November). The water available was compared with a defined threshold to calculate daily values of the sustainability indices.

For maximum production, a medium maturity maize crop, grown in *Kharif* season, requires between 500 and 800 mm of water depending on the climate. The water required to irrigate maize to maturity using the FAO crop factors, is 511.8 mm for 125 growing days (Allen et al., 1998), which was used as the threshold in the model. The main *Rabi* crop is wheat, which for high yields, requires 450–650 mm of water depending on the climate and length of the growing period. In the model simulations a threshold of 470.5 mm of water was used, based on 120 growing days for wheat (Allen et al., 1998).

Due to the randomness in the stochastic generation of rainfall in the model, 30 realisations of the rainfall were used for each land use scenario. The model was run on a daily time step for 12 years and the first 2 years of the output were discarded as a 'warm-up period'. A number of different scenarios were considered based on the area of irrigation, the area of RWH and whether irrigation or RWH is present. The percentage area of RWH land use varied at 0.5, 1, 1.5 and 3% and the irrigated land use percentage area was varied at 5, 10, 15 and 20%. Scenarios were run with and without irrigation and RWH.

In the Arvari River catchment, groundwater supply is not only affected by the irrigation demand, (represented as the area under irrigation) and recharge from RWH, but also by the amount of rainfall. While irrigation area and RWH area are determined by societal management actions, rainfall in the area is an exogenous variable. Drought, defined here as a below average rainfall year, affects each component of the water balance differently, but the analysis here concentrates on the groundwater storage to meet irrigation

demand. To understand the influence of annual rainfall and drought conditions on irrigated agriculture and RWH, the model was run with different numbers of years of below average rainfall. The earlier discussed sustainability indices were compared between each of the drought scenarios. For example the model was run for 12 years; 3 years of above average rainfall, then a below average rainfall period, which ranged from 2 to 6 years, then another 3 years of above average rainfall and so on until the end of the 12 year time period was reached. The first 2 years of the model were again discarded before calculation of sustainability indices.

4. Results

4.1. Sensitivity analysis and adjustment to realistic field values

The results of the sensitivity analysis indicate that the model is highly sensitive to changes of the stochastic rainfall input. This means that variations in rainfall across sub-basins would strongly impact the output distribution, which was also seen in the field where recharge from RWH was clearly affected by rainfall (Glendenning and Vervoort, 2010). The parameters with the highest sensitivity are *CN* and I_a, which for all output had medium to very high sensitivity, because *CN* and I_a determine the partitioning of rainfall into runoff. The most sensitive output is the groundwater level, which was sensitive to each parameter change. Given that this study is focussed on a conceptual comparison, the sensitivity of the model to the groundwater levels is quite important, as this is the key state variable that RWH would influence. However, the high sensitivity would be a problem for making more location specific statements given the high parameter uncertainty. Another sensitive output was *ET*, which is influenced by soil parameters and *LAI*. Evapotranspiration and groundwater levels are also sensitive to the irrigation trigger which influences how much water is transpired, and the amount of water extracted from the groundwater. This sensitivity analysis demonstrates that accurate land use, aquifer properties and soil information would be key to understanding more location specific impacts of RWH than attempted in this study.

Overall, this model displays similar sensitivities to that of a larger parameterised model—SWAT (Lenhart et al., 2002). In that study the most sensitive parameters were *LAI* and *CN*, and physical soil properties such as bulk density, available water capacity and hydraulic conductivity.

In the field the relationship between cumulative rainfall and cumulative potential recharge was non-linear. However this is not fully demonstrated in the model output, which over-estimates recharge for high rainfall. This is also reflected in the percentage of rainfall that becomes recharge, in the model this ranges from 6% in low rainfall years to almost 15% in high rainfall years, which is relatively constant in the field at 7% rainfall (Glendenning and Vervoort, 2010). This demonstrates the difficulties with replicating the RWH structure shape in the model without the addition of many extra parameters that would further increase the uncertainty of the model. Additionally, the impact of rainfall intensity and duration would also have an impact on recharge from RWH structures, which would require longer time series of field data to understand in detail.

Marechal et al. (2006) estimated return flow from irrigated agriculture in a semi-arid groundwater basin in Andhra Pradesh, India, between 69 and 86 mm/year. The study area from that work has much lower *PotET* than the Arvari River catchment, so higher irrigation amounts and therefore more recharge is modelled for Ag2 and Ag3 HRUs, but are similar to the estimates from the regression by Rangarajan and Athavale (2000) (Eq. (13)) (Table 4).

Fig. 2a shows the groundwater table rise at Kaled in well 6 during the monsoon season of 2008. Fig. 2b shows the similar but slower and lower response of the model output. The difference in well response could be because the observed data reflect local groundwater mounding or specific yield is overestimated in the model.

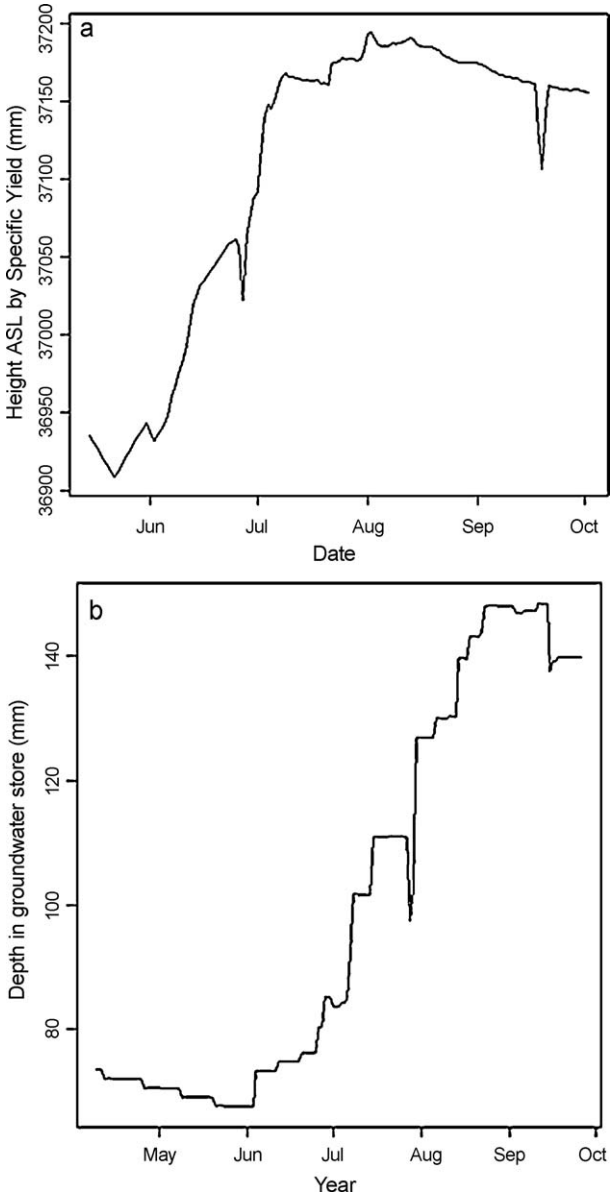


Fig. 2. Groundwater levels in (a) field well KLD6 2008 in village Kaled (mm), (b) from the model output in a similar rainfall year in sub-basin 1.

The final adjusted model output has an average of twenty days of stream flow a year, depending on rainfall, which is similar to field estimates in temporal response (Glendenning and Vervoort, 2010). However the depth of flow in the river is different. The flow depth in the field is much smaller than the predicted stream flow in the model. This could be due to greater transmission losses in the field compared to the regression used in the model.

Overall the results of the adjusted model gave confidence that the model did not give totally unrealistic responses compared to literature studies and the field data. This means that the following sustainability analysis is realistic despite its conceptual nature.

4.2. The system without RWH

The results from the sustainability analysis are presented as box-whisker plots which show the median and 50 and 95% confidence intervals of the distribution. All indices show that as the area of

Table 4
The modelled recharge estimates under different land uses are within the range of the estimates of the natural annual recharge *R* from the precipitation *P* using the regression: $R = 0.147\,P - 6$ (mm/yr) according to Rangarajan and Athavale (2000).

Rainfall (mm/yr)	Natural recharge (mm/yr) from regression	Recharge (mm/yr) from HRUs in the model			
		Commons	Ag1	Ag2	Ag3
426	57	32	23	116	117
748	104	128	100	182	174

irrigation increases, sustainability decreases in the system (Fig. 3). A greater area of irrigation represents a greater crop water demand and therefore more groundwater extraction would occur, so that water levels are below the threshold more frequently. Reliability (*RE*) clearly shows the impact of the increase in irrigation area, which drops from a reliability of 0.8 at 5% irrigation, to half that at 10% irrigated area (Fig. 3a). This difference is not as large from the 15 to 20% irrigated area, because the reliability is already very low at 15%. Resilience (*RS*) again is highest for the smallest irrigated area, but as irrigation area increases, the decrease in the *RS* index is not as much as the *RE* index, because *RS* is already low for a small area of irrigation (Fig. 3b). This indicates that the overall capacity of the system to withstand shocks is minimal. Vulnerability increases with increasing irrigation, and is highest at 20% irrigation, reconfirming the other indices.

4.3. The system with RWH

Introducing RWH creates a more viable water resource system for irrigated agriculture, compared to the previous analysis without RWH. This is indicated by overall higher reliability and resilience, and lower vulnerability (Figs. 4–6).

There is not much difference in reliability as irrigation increases in the *Kharif* and *Rabi* seasons (Fig. 4). In both crop seasons, at 5 and 10% irrigation, the impact of RWH means that these scenarios are almost completely reliable. As RWH area increases, reliability increases, but at 15 and 20% irrigation area, as RWH increases from 1.5 to 3%, reliability does not increase further and slightly decreases, particularly at the 20% irrigation area in the *Rabi* season (Fig. 4). Even though reliability decreases at 15 and 20%, it is still greater than scenarios without RWH (Fig. 3).

At 10% irrigation, in both crop seasons, the resilience index behaves quite differently to the other scenarios (Fig. 5). This is the result of the stepwise irrigation area function in the model, which adjust the fraction of irrigated land to the groundwater level. In the 5% irrigation area scenario the groundwater level is always above the 125 mm threshold, and in the 15 and 20% irrigation area scenarios, the groundwater level is always below the 125 mm threshold. However at the 10% irrigation area scenario, in the *Rabi* season, at 0.5% RWH area, the groundwater levels are greater than 125 mm, so the fraction of irrigation area does not increase and resilience is high. However as RWH area increases, the groundwater levels increase due to increased recharge, and the irrigation area increases, which in turn depletes groundwater levels and thus decreases the resilience index. At 3% RWH the irrigation area function can only increase the irrigation area to maximum 1.2 times or 12%, so extraction cannot increase further, and thus additional RWH increases the resilience. This is similarly reflected in the *Kharif* season at 10% irrigation area, except that the function also affects the 0.5% RWH. Because of the rainfall input in the *Kharif* crop season, recharge increases with increasing RWH.

Despite RWH presence, vulnerability, or the extent of failure, is predictably highest when the irrigation area is largest (20%) (Fig. 6). A larger area of irrigation means a larger volume of groundwater is extracted. Therefore the difference between the required demand and the available water is greater, resulting in a higher vulnerability. At low irrigation area (5%), vulnerability is negligible, but increases with each step in irrigation area. At 15% irrigation, in both crop seasons, as RWH increases from 0.5% RWH to 1% area, vulnerability increases. However from 1.5 to 3% RWH area, the vulnerability stabilises (Fig. 6). At 10% irrigation in the *Kharif* season a similar response was seen in the reliability analysis of the *Rabi* season (Fig. 4), which is the result of the increase in irrigation area.

Compared to the first analysis with irrigation and no RWH (Fig. 3), the addition of RWH in the system increases reliability, resilience and decreases the vulnerability of the water resource system for irrigated agriculture. Within an individual irrigation scenario, as RWH area doubles from 1.5 to 3%, the benefit to the system is not large and in some cases the sustainability indices actually decrease. This suggests that there is a limit in the area of RWH that gives a maximum recharge benefit and beyond which the benefit is marginal. In addition, the fact that farmers respond to water levels and increase their irrigation area means that some levels of RWH area are less sustainable depending on management practices.

4.4. The influence of rainfall

To understand the influence of variations in annual rainfall on irrigated agriculture and RWH, the indices were compared between below and above average annual rainfall years. As discussed, the long-term average from the Thanagazi station data of 705.8 mm was used to separate the wet and dry rainfall years. The analysis concentrated on the *Rabi* season, as the *Kharif* season displays a similar response. As the *Rabi* season has less daily rainfall, the impact of recharge from RWH would be the primary influence on whether the threshold demand is met.

In all indices, the above average rainfall years show higher sustainability than below average rainfall years. Systems with and without RWH have higher reliability with above average rainfall. However when rainfall is below average, the reliability of the system with RWH is greater than a system without RWH. This is clearly demonstrated in the 5 and 10% irrigation scenarios, where the system has almost 100% reliability despite below average rainfall (Fig. 7). The reliability of a system without RWH has median reliability of 80% for 5% irrigation area and 20% for 10% irrigation area (Fig. 7). When rainfall is above average, RWH in a system gives 100% reliability up to the 15% irrigation scenario. In the below average rainfall scenarios, reliability decreases slightly when increasing RWH from 1.5 to 3% (Fig. 7).

Without RWH, resilience (*RS*) is lower in below average rainfall years than above average rainfall years. But a system with RWH has better resilience than a system without RWH. With RWH, in the below average rainfall scenario, *RS* cannot be measured at 5% irrigation scenario (Fig. 8). At 15 and 20% irrigation, as RWH increases from 1.5 to 3% no large improvements in the index value is seen. At 10% irrigation in the below average rainfall with RWH, the results are similar to Fig. 5, which is the result of the irrigation area function operating. Groundwater levels are low at 0.5% RWH so the irrigation area does not increase, and resilience is high. From here, as RWH area increases, resilience decreases because groundwater levels rise and the effect of the irrigation area function increases. With RWH in the system and above average rainfall, resilience cannot be measured for 5, 10 and 15% irrigation scenarios (Fig. 8). At 20% irrigation the irrigation area function is triggered. But at 3%

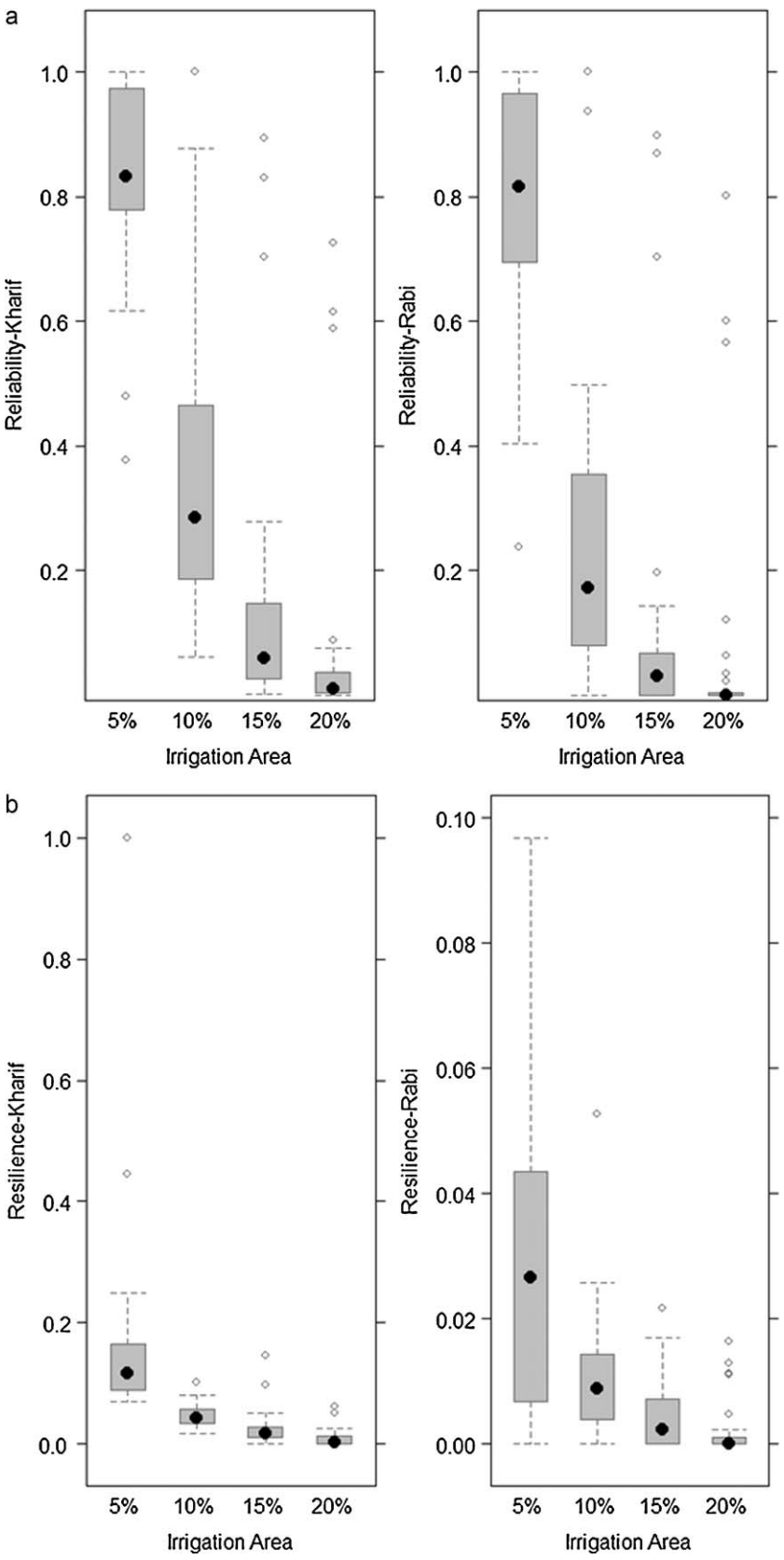


Fig. 3. Distribution of the sustainability indices, with median and 50 and 95% confidence intervals of the distribution, for different areas of irrigation without RWH for *Kharif* and *Rabi* (a) Reliability Index, (b) Resilience Index.

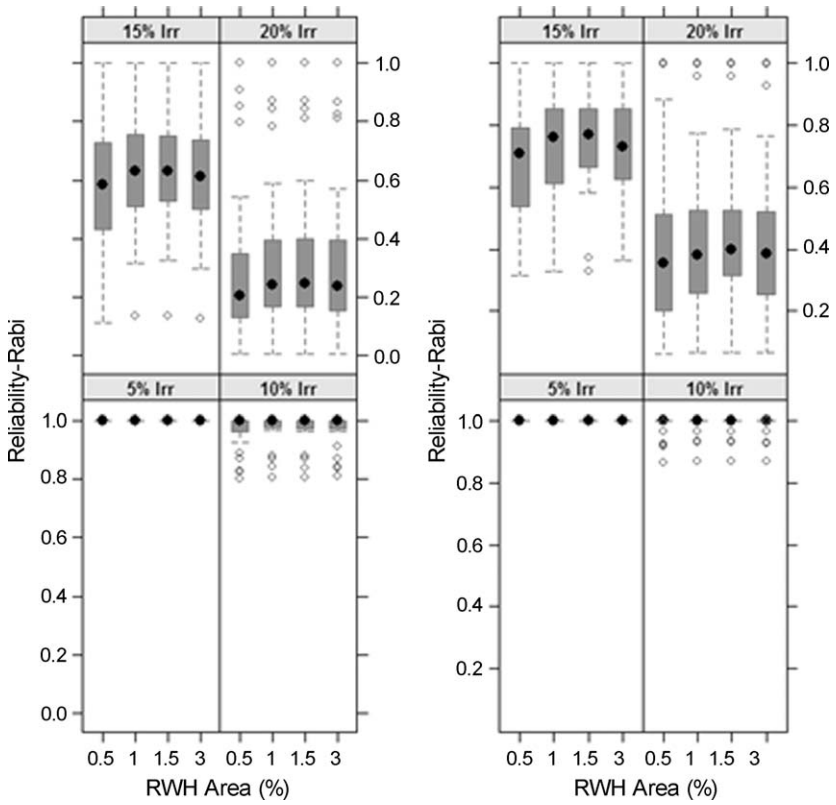


Fig. 4. Distribution of the reliability index, with median and 50 and 95% confidence intervals of the distribution, for different areas of irrigation and RWH for *Kharif* and *Rabi* season.

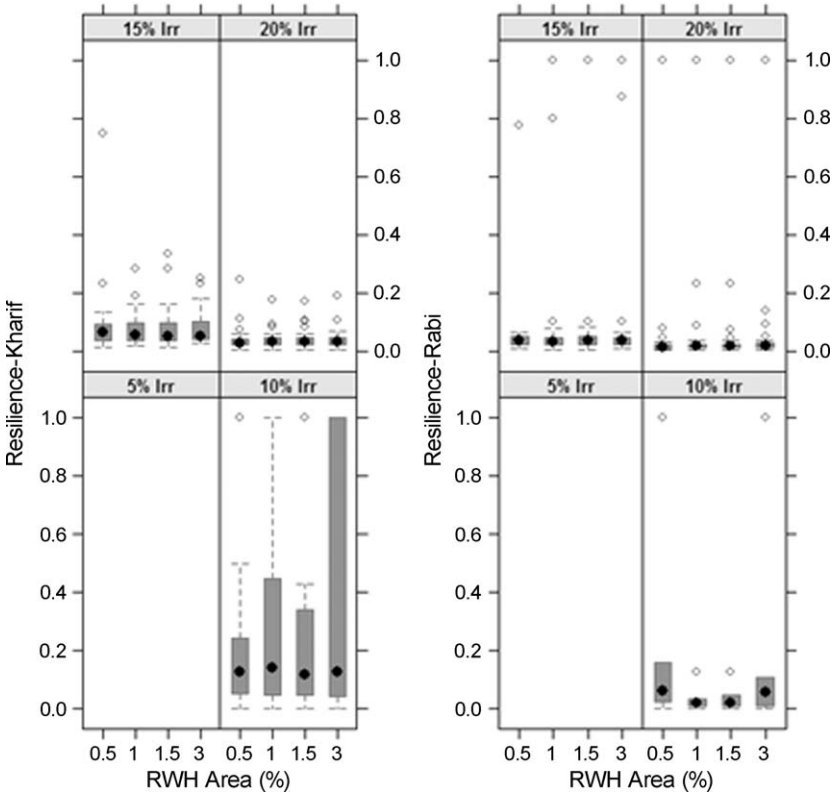


Fig. 5. Distribution of the resilience index, with median and 50 and 95% confidence intervals of the distribution, for different areas of irrigation and RWH for *Kharif* and *Rabi* season.

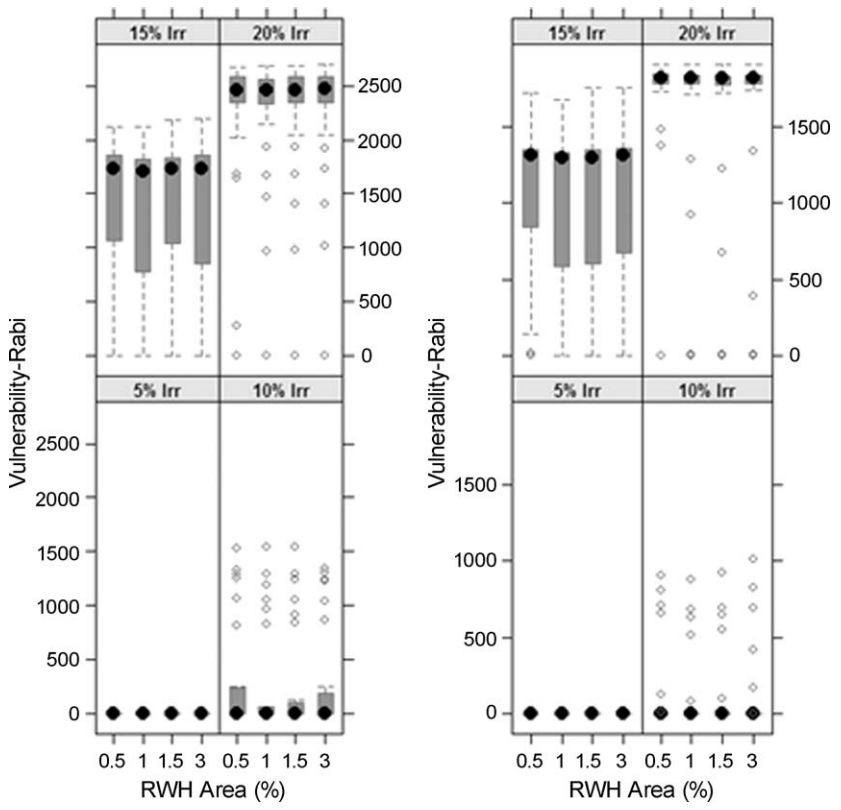


Fig. 6. Distribution of the vulnerability index, with median and 50 and 95% confidence intervals of the distribution, for different areas of irrigation and RWH for *Kharif* and *Rabi* season.

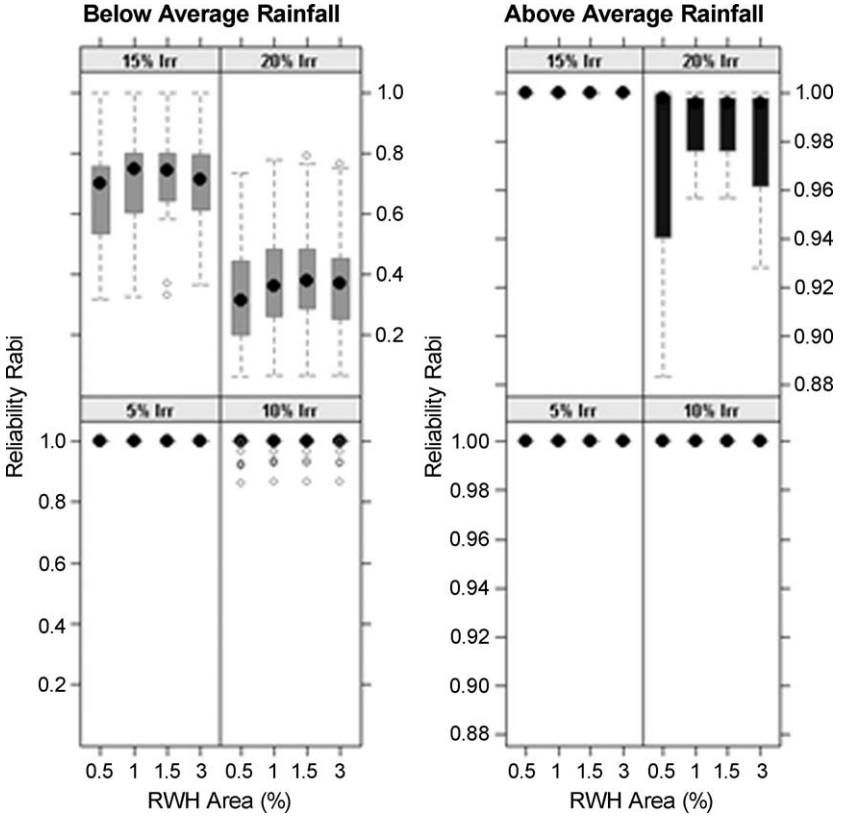


Fig. 7. Reliability in *Rabi* with below average rainfall and above average rainfall with various scenarios of RWH and irrigated agriculture, with median and 50 and 95% confidence intervals of the distribution.

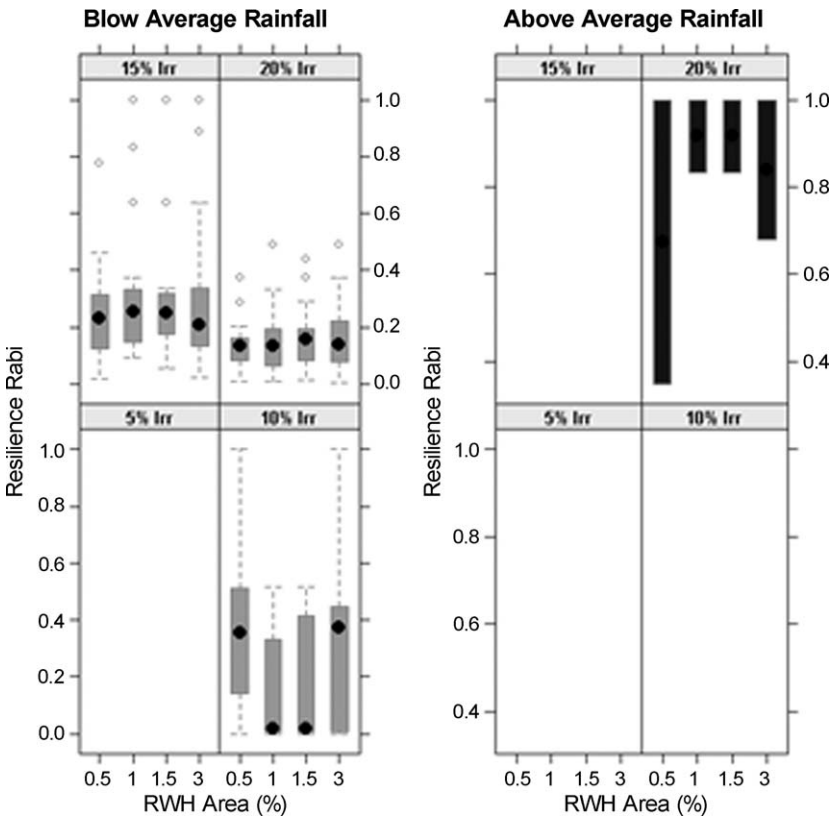


Fig. 8. Resilience in *Rabi* with below average rainfall and above average rainfall with various scenarios of RWH and irrigated agriculture, with median and 50 and 95% confidence intervals of the distribution.

RWH area, the groundwater levels are not high enough to activate the irrigation area function, and so resilience is lower.

The vulnerability index shows that a system with RWH is less vulnerable than a system without RWH in above and below average rainfall years. As RWH area increases, vulnerability is similar for all levels of RWH area.

When RWH is present with below average rainfall, the sustainability indices are not as positive as when rainfall is above average, due to lower recharge from RWH. However in below average rainfall years, the indices are greater with RWH in the system than without RWH. This means RWH alleviates some of the deficit in below average rainfall years, even though recharge would be less than in an above average rainfall year.

4.5. Drought conditions and RWH impact

RWH provides within season buffering, but may not provide longer term supply if drought occurs, because the local pumping practices for irrigation reduces the buffer between seasons and recharge from RWH would decrease. However, in the simulation analysis, the overall sustainability indices, with increasing series of concurrent below average rainfall years, did not show a strong difference as the drought period increases. This suggests that RWH is able to provide some inter annual buffer in low rainfall periods. Overall the resilience index was low, but resilience and reliability decreased as the drought period increases. These changes are smaller than without RWH in the system.

4.6. Stream flow impacts of RWH

The slope of the rainfall–stream flow relationship was used to assess the impact of RWH on stream flow. The slope decreases with increasing irrigation. Stream flow still occurs at low annual rainfall (Table 5), but the large stream flow events decrease. This is because the model converts the commons HRU, with a higher *CN*, to irrigated HRUs, which has lower runoff. Therefore more water is 'captured' through irrigated land use, and 'blue' water (rivers and aquifers) is converted into 'green' water (soil water and plant water use) (Falkenmark, 2003).

When RWH is introduced the slope of the rainfall–stream flow relationship increases, but the intercept with the rainfall axis shifts from 400 to 500 mm. This indicates that larger rainfall events are needed to generate stream flow. In the model, RWH captures runoff that would otherwise reach the stream and so the amount and frequency of water entering the stream decreases. Smaller rainfall events are captured, but larger events overflow from the RWH structures and move into the river. If the RWH area is further increased, the number of stream flow events decreases further, but the greatest change in stream flow occurs from 0% RWH area to 0.5% RWH area (Table 5). As a result the mean and maximum runoff coefficient decreases sharply with the introduction of RWH.

Table 5
Analysis of stream flow with different scenarios of RWH and irrigation.

Scenario		Linear model		Stream flow as % rainfall		
RWH % Area	Irrigation % area	Slope	x axis intercept	Mean	Standard deviation	Maximum
0	0	0.017	194	7.0	2.7	19.6
0	15	0.011	118	5.4	1.7	16.5
0.5	15	0.021	486	2.9	2.8	18.3
1	15	0.021	524	2.0	2.7	17.0
1.5	15	0.021	524	1.6	2.7	16.3
3	5	0.024	422	4.4	3.1	17.4
3	10	0.023	513	2.2	2.9	16.1
3	15	0.019	526	1.2	2.4	14.9

5. Discussion

This study presents a conceptual model designed to allow relative sustainability comparisons between different RWH scenarios. Despite the aim to keep the parameters in the model low, the field results indicated that some complexity and variability was necessary to capture all the behaviour. As such the model has 23 parameters (Table 3). Compared to a low parameter model such as IHACRES (Evans and Jakeman, 1998) it is complex. However, low parameter models, while effective for modelling stream flow output, do not allow investigation of catchment internal hydrological processes necessary for the RWH sustainability analysis. Furthermore, existing models such as the TEDI model (Lowe et al., 2005) and the HYLUC-Cascade model (Bishop et al., 2009) represent similar hydrological processes to estimate the impact of farm dams on catchment flows, but do not incorporate any surface-groundwater interactions. Those models that do include a reservoir model and surface water–groundwater interactions, such as SWAT (Arnold and Fohrer, 2005), are far too complex (more than 100 parameters) and data intensive. The semi-distributed conceptual model built here is similar in structure to models like SWAT. Because the model focuses on conceptual relationships and there is limited data, uncertainty estimation is not possible and true field representation is also not possible.

At present agriculture in India is far more reliant on groundwater than the earlier rain-fed systems. As a result of this dependence, the groundwater resource has become over-exploited in some areas, so that the groundwater buffer, which provides resilience against shocks such as drought, may no longer exist (Moench, 2007). Agricultural drought occurs widely in India, mostly in the arid and semi-arid areas, lowering groundwater levels (Mudrakartha, 2007). Increased recharge through RWH is seen as an important element for providing some resilience against shocks such as drought in a groundwater irrigated agriculture system.

Under the RWH scenarios in the model, irrigated agriculture is more viable than in a system without RWH. However the analysis also showed that continuously increasing RWH area does not bring additional benefits. Instead as RWH area increases, the system reaches a limiting point, where the sustainability indices do not increase further, and in some cases decrease. This is related to several issues. Firstly because the runoff from a catchment area is finite and more RWH means that water is just spread across more storage area. As a result, the depth in each of the RWH structures is smaller and the recharge per structure decreases. Recharge also cannot increase beyond a certain limit for each structure. Secondly, the aquifer storage capacity is limited. If the area of irrigation is large (in this analysis up to 20% of the catchment area), RWH many not be able to reduce the stress on the water resource system, because demand is too large. Finally, the modelled farmer response is to increase irrigation area with increased water availability.

The results from this conceptual analysis can provide policy and management guidelines for NGOs, other agencies and communities that invest in RWH, because the benefit to irrigated agriculture may not be worth the cost at high levels of RWH development in a catchment. A greater understanding of this limiting capacity could aid in RWH development programs. This reinforces theoretical speculations about catchment-scale impacts of RWH by Kumar et al. (2006). Their study concluded that a greater degree of RWH development would decrease the social, economic and environmental marginal benefits of building additional RWH structures. This means that if there are already many RWH structures in a catchment, the marginal benefit for every new RWH structure is smaller than for structures already built, and the marginal cost is greater due to the social and environmental costs of harvesting every unit of water and potential for over-appropriation. The marginal benefit at higher RWH development decreases due to a number of reasons including low aquifer storage capacity and lower chances of finding appropriate sites for RWH development (Kumar et al., 2006). In the conceptual model in this study, the assumed small aquifer storage capacity could explain why increasing RWH area does not strongly increase the benefit to the groundwater system. In addition, the nature of runoff and storage into RWH structures could also reduce the benefits of recharge from RWH. Before further recommendations, these results must be compared with other models that have been calibrated and validated to a specific catchment, based on more extensive field data.

Despite these limitations, the analysis from the model confirms that RWH increases the viability of groundwater irrigated agriculture if short-term mild drought conditions occur. However, in the model longer term drought does not seem to be alleviated by RWH. Kumar et al. (2006) argue that RWH has an extremely limited potential to provide reliable water supply, because of high rainfall variability and high *PotET*. In contrast, the conceptual analysis here shows that while the sustainability indices decrease when drought occurs, the modelled system functions better with RWH than without.

Another important aspect of RWH impacts are the hydrological changes in a catchment, which were examined by comparing stream flow changes under different scenarios. When RWH is introduced, stream flow strongly decreases. Capturing local runoff upstream in RWH storages addresses problems of frequent drought and widespread poverty in upper catchments. However the 'blue' water investments, like irrigation canals, are generally located downstream (for example Sainthal Sagar Dam in the Arvari River catchment) and depend on large volumes of runoff (Sakthivadivel, 2008). As RWH leads to an increase in 'green' water used for irrigated agriculture upstream, this will affect 'blue' water availability for downstream users, including irrigators and ecosystems, because flows decrease. However RWH could also have positive environmental impacts as a result of reduced land degradation. For example in the Arvari River catchment, more area has been re-forested since RWH development as the community recognises the importance of erosion control. There could also be improvements in water quality as runoff is slowed through the catchment, so larger sediments would filter out and the erosive power of the flows is reduced. Finally longer residence times increases possible feedbacks between water and vegetation and could lead to improved environmental condition. This would require further field analysis.

The scenarios of irrigation area and RWH area highlight the link and strong feedbacks between irrigation area and RWH. The conceptual model does not fully capture the changing dynamics of the irrigation area. Realistically, if groundwater levels were low, farmers might also reduce the area of irrigation, alleviating pressure on the groundwater resource. While the model results show that RWH has a positive impact for the sustainability of irrigated agriculture, it also shows that if the supply of groundwater increases due to RWH, then local demands increase requiring demand-side management of groundwater.

6. Conclusions

As pumping technology has changed, groundwater use has increased the accessibility of a reliable source of irrigation water for small-holder farmers across India, alleviating poverty in many rural areas. Yet the presence of millions of private wells has meant that groundwater is over-extracted and water levels are declining. As a result RWH has been widely implemented to raise falling groundwater levels, but with no scientific or hydrological studies to confirm or aid in implementation.

This study has examined catchment-scale RWH impacts using a conceptual water balance model. The simulation results show

that RWH has a positive impact on groundwater recharge and sustainability of irrigated agriculture, but decreases stream flow downstream. Rainwater harvesting specifically changes 'blue' water into 'green' water as the irrigation area increases as RWH is implemented. However, the resilience and sustainability of the groundwater system increases with limited levels of RWH.

The heavy reliance on groundwater in many parts of India means that the RWH impacts must be examined further for effective catchment development programs in India, and the results should be conveyed to the groundwater users themselves. RWH is a supply-side groundwater management strategy, so demand-side groundwater management strategies should also be considered. Future work is needed to examine the conclusion drawn from the simulation study, that increasing RWH *ad infinitum* does not increase strongly benefit for irrigation.

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